## Question

The ground-state atomic carbon can react with alkenes. When *cis*-2-butene is used, the reaction produces two different pairs of racemates, $\mathbf{A}$ and $\mathbf{B}$, with a product ratio $\mathbf{A:B}$ close to $1:1$. If *trans*-2-butene is used, two pairs of racemates, $\mathbf{B}$ and $\mathbf{D}$, and one achiral product, $\mathbf{C}$, are formed. The ratio of $\mathbf{B:C}$ is approximately $1:1$, while the yield of $\mathbf{D}$ is relatively low. Indicate the symmetry of each product $\mathbf{A,B,C,D}$.

A. $\mathbf{A}:C_2,\mathbf{B}:C_1,\mathbf{C}:S_4,\mathbf{D}:D_2.$

B. $\mathbf{A}:C_1,\mathbf{B}:D_2,\mathbf{C}:C_{2\mathrm{h}},\mathbf{D}:C_2.$

C. $\mathbf{A}:C_2,\mathbf{B}:C_1,\mathbf{C}:S_4,\mathbf{D}:D_2.$

D. $\mathbf{A}:C_2,\mathbf{B}:D_2,\mathbf{C}:C_{2\mathrm{v}},\mathbf{D}:C_1.$

E. $\mathbf{A}:D_2,\mathbf{B}:C_1,\mathbf{C}:C_{2\mathrm{v}},\mathbf{D}:C_2.$

F. $\mathbf{A}:D_2,\mathbf{B}:C_2,\mathbf{C}:C_{2\mathrm{h}},\mathbf{D}:C_1.$

## Answer

**Correct Answer: C**

## Detailed Explanation

The reaction of ground-state carbon atoms with alkenes yields products featuring a spiro[2.2]pentane skeleton.

**CHECKPOINT** 1 PTS

Product has spiro[2.2]pentane skeleton

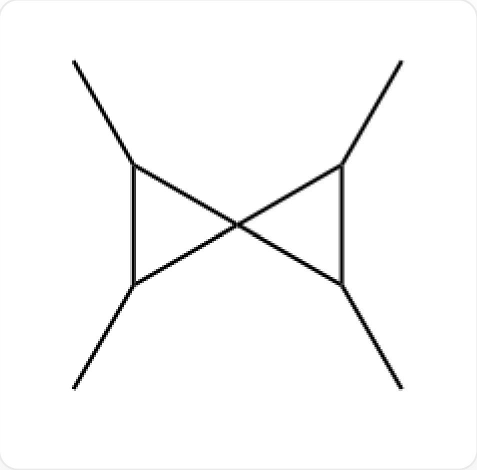
For *cis*-2-butene or *trans*-2-butene, the resulting product should incorporate four methyl groups on the non-spiro atoms of the spiro[2.2]pentane molecule.

**CHECKPOINT** 1 PTS

Product has four methyl groups

Thus, ignoring stereoisomers, the product is:



CC1C(C)C12C(C)C2C

The spectroscopic term for the ground-state carbon atom is $^3P$, and its reactivity resembles that of a triplet carbene.

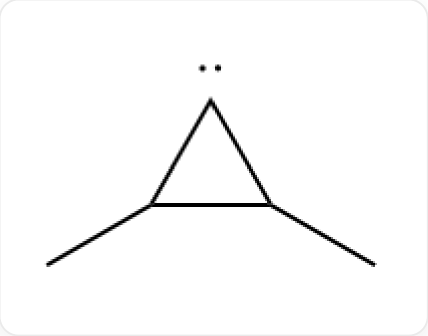
**CHECKPOINT** 1 PTS

Ground-state carbon atom resembles triplet carbene

Consequently, its reaction mechanism with the first molecule of *cis*-2-butene or *trans*-2-butene involves radical addition, forming a diradical intermediate, followed by spin inversion and cyclization to yield a cyclopropane carbene intermediate.

**CHECKPOINT** 1 PTS

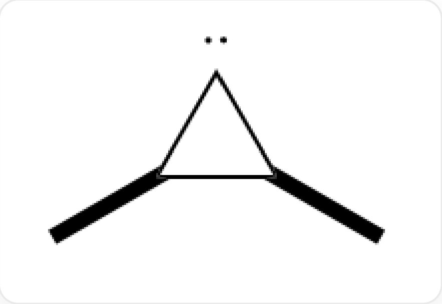
First step is radical addition



CC1[C..]C1C

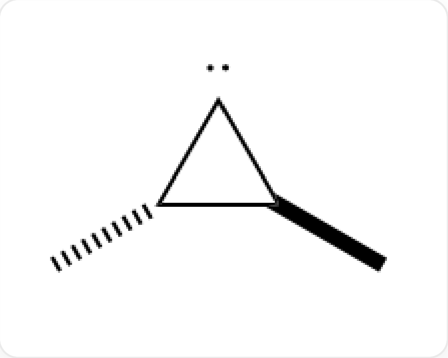
The generated cyclopropane intermediate exists in both *cis* and *trans* configurations, with their abundance ratio close to $1:1$.

The *cis* intermediate is:



C[C@H]1[C..][C@@H]1C

The *trans* intermediate is:



C[C@H]1[C..][C@H]1C

and its enantiomer.

**CHECKPOINT** 1 PTS

Intermediate exists in both *cis* and *trans* forms

The carbene intermediate, influenced by the high ring strain of the cyclopropane, adopts a singlet state, with the HOMO orbital lying within the plane of the cyclopropane ring and the LUMO orbital perpendicular to it.

**CHECKPOINT** 1 PTS

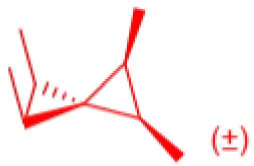
Cyclopropane intermediate is a singlet

This singlet intermediate further undergoes a $[1+2]$ cycloaddition reaction with the second molecule of *cis*-2-butene or *trans*-2-butene, yielding stereospecific products.

**CHECKPOINT** 1 PTS

$[1+2]$ cycloaddition occurs

If *cis*-2-butene reacts with the *cis* intermediate, the product is:



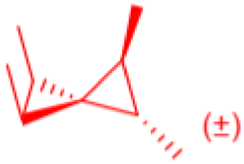
C[C@@H]1[C@H](C)[C@]12[C@@H](C)[C@H]2C

This product contains one $R$ and one $S$ chiral carbon in each cyclopropane ring, along with a chiral axis ( $R$ or $S$ ). It can only be formed from the reaction of ground-state carbon atoms with *cis*-2-butene, designated as $\mathbf{A}$ , with a point group of $C_2$ .

**CHECKPOINT** 1 PTS

$\mathbf{A}$ has point group $C_2$

If *cis*-2-butene reacts with the *trans* intermediate or *trans*-2-butene reacts with the *cis* intermediate, the product is:



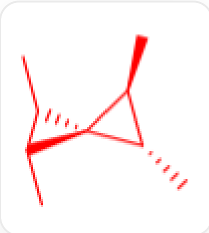
C[C@@H]1[C@H](C)C12[C@@H](C)[C@@H]2C

This product lacks axial chirality but exhibits asymmetric chiral center distribution: one cyclopropane ring has one $R$ and one $S$ chiral carbon, while the other has two $R$ or two $S$ chiral carbons. This molecule can be formed from either *cis*-2-butene or *trans*-2-butene, designated as $\mathbf{B}$ , with a point group of $C_1$ .

**CHECKPOINT** 1 PTS

**B** has point group $C_1$

If *trans*-2-butene reacts with the *trans* intermediate, two outcomes are possible. Under lower steric hindrance, the product is:



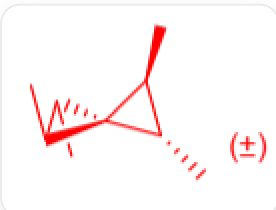
C[C@H]1[C@H](C)C12[C@@H](C)[C@@H]2C

Here, one cyclopropane ring has two $R$ chiral carbons, and the other has two $S$ chiral carbons, resulting in an achiral molecule with an $S_4$ point group, designated as $\mathbf{C}$.

**CHECKPOINT** 1 PTS

**C** has point group $S_4$

Under higher steric hindrance, the product is:




C[C@@H]1[C@@H](C)C12[C@@H](C)[C@@H]2C

Here, all four tertiary carbons are either $R$ or $S$, and the molecule lacks axial chirality, with a point group of $D_2$, designated as $\mathbf{D}$.

**CHECKPOINT** 1 PTS

$\mathbf{D}$ has point group $D_2$

**CHECKPOINT** 1 PTS

Yield difference between $\mathbf{C}$ and $\mathbf{D}$ arises from steric hindrance